NYSC Directed Study Notes

Relativity

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Contents

1 Day 1: Special Relativity & Linear Algebra **2**
- 1.1 Electricity and Magnetism . . . 2
- 1.2 Galilean Transformations and Their Problems . . . 3
- 1.3 Postulates of Special Relativity . . . 4
- 1.4 3 Consequences of Postulates . . . 4
  - 1.4.1 Relativity of Simultaneity . . . 4
  - 1.4.2 Time Dilation . . . 5
  - 1.4.3 Length Contraction . . . 7
- 1.5 Lorentz Transformations . . . 8
- 1.6 Linear Algebra Primer . . . 10
  - 1.6.1 Dot and Cross Products . . . 10
  - 1.6.2 Basis Vectors . . . 11
  - 1.6.3 Orthonormal Basis Vectors . . . 12
  - 1.6.4 Matrices . . . 12
  - 1.6.5 Summary . . . 14

2 Day 2: General Relativity **16**
- 2.1 Lorentz Transformations in Matrix Form . . . 16
- 2.2 Newtonian Gravity . . . 19
- 2.3 Transition to General Relativity . . . 21
- 2.4 General Relativity . . . 21
- 2.5 Schwarzschild Metric . . . 24

3 Day 3: Applications of and Problems with General Relativity **25**

1 Day 1: Special Relativity & Linear Algebra

Question 1. *What do people know about relativity? What do they think it's about? What do they hope to learn? Can they distinguish special and general relativity*

Today, we will:

1. Discuss how the constancy of the speed of light brought about troubles in classical physics
2. How Einstein solved these problems with *special* relativity
3. The unintuitive consequences of special relativity
4. Begin moving towards general relativity by introducing concepts important to special and general relativity and necessary mathematical basics

1.1 Electricity and Magnetism

- Maxwell's Equations were collected in 1865 by James Clerk Maxwell
  - the laws were mostly already discovered individually... Maxwell was the first to combine them in a coherent theory of electromagnetism)

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0} \quad \nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$$

$$\nabla \cdot \boldsymbol{B} = 0 \quad \nabla \times \boldsymbol{B} = \mu_0 \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} + \mu_0 \boldsymbol{J}$$

These equations result in a wave equationtell us that the speed of light is fixed.

Activity 1. *Finding a Constant Speed of Light in Maxwell's Equations*

First using a gradient identity:

$$\begin{aligned} \nabla \times (\nabla \times \boldsymbol{E}) &= \nabla(\nabla \cdot \boldsymbol{E}) - \nabla^2 \boldsymbol{E} \\ &= \nabla\left(\frac{\rho}{\epsilon_0}\right) - \nabla^2 \boldsymbol{E} = -\nabla^2 \boldsymbol{E} \end{aligned} \tag{1}$$

Then using Maxwell's equations:

$$\begin{aligned}\nabla \times (\nabla \times \boldsymbol{E}) &= \nabla \times (-\frac{\partial \boldsymbol{B}}{\partial t}) = -\frac{\partial}{\partial t}(\nabla \times \boldsymbol{B}) \\ &= -\frac{\partial}{\partial t}\left(\mu_0\epsilon_0\frac{\partial \boldsymbol{E}}{\partial t} + \mu_0\boldsymbol{J}\right) \\ &= -\mu_0\epsilon_0\frac{\partial^2 \boldsymbol{E}}{\partial t^2} - \mu_0\frac{\partial \boldsymbol{J}}{\partial t}\end{aligned} \tag{2}$$

These combined, using the fact that $\partial_t \boldsymbol{J} = 0$ tell us that

$$\nabla^2 \boldsymbol{E} = \mu_0\epsilon_0\frac{\partial^2 \boldsymbol{E}}{\partial t^2} \tag{3}$$

Notice this is a wave equation for the electric field, moving with a speed

$$\frac{1}{c^2} = \mu_0\epsilon_0 \implies \boxed{c = \frac{1}{\sqrt{\mu_0\epsilon_0}}} \tag{4}$$

Question: What units and value does c have?

- Maxwell's equations, and all of physics, are frame-independent. This tells us that c should always be 3×10^8 m/s in *any* frame of reference.

1.2 Galilean Transformations and Their Problems

- Galilean transformations are ways of transferring between references frames and the addition of velocities that is very intuitive from our classical picture of the world.

  Activity 2. *Consider a raft moving down a river at speed v when a cannon is fired from it. The raft's captain sees the cannonball fired at a speed u. What speed u will an observer on the riverbank see the cannonball moving at?*

  $$u = u' + v \tag{5}$$

  Where u' is the speed of the raft.

- If velocities are always added when transforming between reference frames but c is always constant, how can we resolve this problem?

- For example, instead consider a laser beam being emitted from a spaceship moving near the speed of light. What speed would a stationary observer see the light traveling at?

- In 1905, Einstein derived a formula for this problem, which included a correction factor to the Galilean transformation:

$$u' = \frac{v + u}{1 + \frac{vu}{c^2}} \tag{6}$$

Question 2. *What does u' look like for small u and v? What does it look like in the limit where u & $v \to c$?*

1.3 Postulates of Special Relativity

- While working a patent clerk office job in 1905, Einstein worked on resolving this tension between c and Galilean physics in his spare time.

- Based upon theoretical considerations and experimental evidence, Einstein began resolving this tension by taking as physical principles:

- In **inertial frames of reference** (ones with no acceleration)

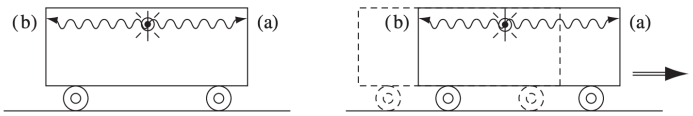
1. Physics is the same
   - There are no "preferred reference frames" or preferred observers
2. Speed of light is universally c

1.4 3 Consequences of Postulates

- Directly from these postulates, Einstein was able to arrive at three monumental discoveries:

1.4.1 Relativity of Simultaneity

- Events that are simultaneous in one frame of reference may not be so in another frame:



(Introduce our notation for $\mathcal{S}$ denoting the rest frame and $\mathcal{S}'$ as the moving inertial system)

1.4.2 Time Dilation

- First of all,

  Question 3. *How do we measure time?*

- In light clocks, we measure a second as the time it takes for a light beam to "tick" once:

$$t' = \frac{\ell}{c} \tag{7}$$

  If the light clock is moving at a speed v in an inertial frame $\mathcal{S}'$ relative to a stationary observer $\mathcal{S}$, we do not get the same form.

  Activity 3. *With the resulting triangular path of the light beam, what time does the stationary observer see pass?*

  Answer:

$$t = \frac{\sqrt{\ell^2 + (vt)^2}}{c} \tag{8}$$

  Activity 4. *Now solve Eq.* (8) *for* t.

  Answer:

$$t = \frac{\ell}{c\sqrt{1 - \left(\frac{v}{c}\right)^2}} \tag{9}$$

- Notice that we can substitute t into this expression for t' to give us

$$t = \frac{t'}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \tag{10}$$

- We then define the **Lorentz factor**:

$$\gamma := \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \tag{11}$$

so that

$$\begin{aligned} t &= \gamma t' \\ \implies \boxed{t' = \frac{t}{\gamma}} \end{aligned} \tag{12}$$

Question 4. *What does this result tell us about how time is experienced in the moving frame of reference $\mathcal{S}'$?*

Answer: ***Moving clocks run slow!***

- It is useful to examine the form of γ in more detail and use a trick ubiquitous in physics.

Question 5. *What does γ look like when plotted as a function of v?*

Hint*: The low-order Taylor expansion for γ looks like*

$$\frac{1}{\sqrt{1 - x^2}} \approx 1 + \frac{x^2}{2} + \frac{3x^4}{8} + \dots \tag{13}$$

Answer*: as $v \to c \implies \gamma \to \infty$*

- This leads to the famous twin paradox:

Activity 5. *Imagine two 20 year old twins, Alice and Bob. One of the twins, Alice takes off on a spaceship which travels at .9c to another planet. In total, 10 years elapses on Alice's watch during her journey. How old is Alice when she returns?*

Hint: *Consider how much time t Bob experiences when Alice experiences 10 years in her moving frame of reference $\mathcal{S}'$.*

Solution: *We must solve for t:*

$$t = \gamma t' = \frac{10}{\sqrt{1 - \left(\frac{.9c}{c}\right)^2}} \approx 22.94 \tag{14}$$

This tells us that when Alice returns, Bob is 42 years old while Alice is only 30 years old!

1.4.3 Length Contraction

- Now what if we measured the length of an object, say a train cab (as before), using light pulses? Does special relativity influence spatial measurements as well?

- (first draw the cart diagram on the board with Δx_1 and Δx_2 lengths)

  Activity 6. *Derive the total time Δt a stationary observer would see a light beam passing from one end to the other in a train cab moving at a speed v.*

  Answer:

$$\begin{aligned} \Delta t'_{tot} &= \frac{2\Delta x'}{c} \\ \Delta t_1 &= \frac{\Delta x}{c+v} \\ \&\ \Delta t_2 &= \frac{\Delta x}{c-v} \\ \implies \Delta t_{tot} &= \frac{2\Delta x}{c\left(1-\left(\frac{v}{c}\right)^2\right)} \end{aligned} \tag{15}$$

  Inserting the expression for $\Delta t = \gamma \Delta t'$:

$$\gamma \Delta t' = \frac{2\Delta x}{c}\gamma^2 \tag{16}$$

  Then using our derived expression for $\Delta t'$, we get

$$\begin{aligned} \frac{2\Delta x'}{c}\gamma &= \frac{2\Delta x}{c}\gamma^2 \\ \implies \boxed{\Delta x' = \gamma \Delta x} \end{aligned} \tag{17}$$

  Question 6. *What does this expression tell us about the length of objects as they move at high speeds?*

  Answer: As $v \uparrow \implies \Delta x \downarrow$ (as an object moves at greater speeds, a stationary observer of it sees the length of that object decrease)

- Note that we have let v be in the $+\Delta x$ direction. This tells us that **lengths are always contracted along the direction of motion**
  - Lengths perpendicular to the direction of motion *are not* contracted

1.5 Lorentz Transformations

- Back to Galilean transformations:
  - Imagine an observer moving with a speed v to the right in a train seeing a stationary observer throw a ball right. The stationary observer records the ball traveling a distance Δx in a time t.

    Question 7. *How far would the moving observer see the ball move?*

    Answer: *the moving observer sees the stationary observer moving backwards with a speed v to the left, which means*

    $$\Delta x' = \Delta x - vt \tag{18}$$

  - Since we assume that the flow of time is the same for al observers, we get the transformations

    $$\begin{cases} t' &= t \\ x' &= x - vt \\ y' &= y \\ z' &= z \end{cases} \tag{19}$$

    For an observer moving in $+x$ with speed v.
  - These are the **Galilean transformations**
  - But we just derived that

    $$x' = \gamma x \tag{20}$$

    when speeds reach c. Substituting this expression in, we get

    $$\boxed{x' = \gamma(x - vt)} \tag{21}$$

  - Also since

    $$x = x' + vt' \tag{22}$$

    in Galilean relativity (again, the moving observer sees the stationary observer moving in the $-x$ direction), we also have

    $$x = \gamma(x' + vt') \tag{23}$$

Activity 7. *Eliminate the x' term in the x expression to get an expression for t'.*

Answer:

$$\begin{aligned} x' = \gamma(x - vt) \quad \& \quad x &= \gamma(x' + vt') \\ \implies x &= \gamma(\gamma(x - vt) + vt') \\ \implies x &= \gamma^2 x - \gamma^2 vt + \gamma vt \\ \implies \ldots \frac{(1-\gamma^2)x + \gamma^2 vt}{\gamma v} &= t' \\ \implies t' &= \frac{(1-\gamma^2)x}{\gamma v} + \gamma t \end{aligned} \tag{24}$$

and this simplifies down to

$$\boxed{t' = \gamma\big(t - \frac{vx}{c^2}\big)} \tag{25}$$

Our derived transformations that take the constancy of the speed of light into account,

$$\begin{cases} t' &= \gamma\big(t - \frac{vx}{c^2}\big) \\ x' &= \gamma(x - vt) \\ y' &= y \\ z' &= z \end{cases} \tag{26}$$

are called **Lorentz transformations**, which Einstein used to replace Galilean transformations.

Activity 8. *Remember the relativistic velocity addition formula that we discussed much earlier to motivate why relativity was needed? We are now in the position to derive it.*

In the stationary frame,

$$u := \frac{dx}{dt}. \tag{27}$$

But in the moving ("primed") frame, we have

$$u' = \frac{dx'}{dt'}. \tag{28}$$

Use the differential form of the Lorentz transformation:

$$\begin{aligned} x' = \gamma(x - vt) &\implies dx' = \gamma(dx - vdt) \\ t' = \gamma\big(t - \frac{v}{c^2}x\big) &\implies dt' = \gamma\big(dt - \frac{v}{c^2}dx\big) \end{aligned} \tag{29}$$

We then have that

$$\begin{aligned} u' &= \frac{\gamma(dx - vdt)}{\gamma\big(dt - \frac{v}{c^2}dx\big)} \\ &= \frac{dx/dt - v}{1 - v/c^2 dx/dt} = \boxed{\frac{u - v}{1 - \frac{uv}{c^2}} = u'}. \end{aligned} \tag{30}$$

- To begin writing Lorentz transformations in a more concise notation and begin introducing the underlying mathematical concept of space-time that will allow us to transition to general relativity, we must first cover linear algebra.

1.6 Linear Algebra Primer

1.6.1 Dot and Cross Products

Question 8. *Does anyone know/remember what a vector is defined as? What are some examples? What are* not *vectors?*

- A **vector** is any quantity with magnitude and direction

$$\vec{V} = \langle V_x, V_y, V_z \rangle \tag{31}$$

Question 9. *Does anyone know/remember the two vector operations?*

Dot Product:

$$\vec{V} \cdot \vec{W} := V_x W_x + V_y W_y + V_z + W_z \tag{32}$$

Also

$$\vec{V} \cdot \vec{W} = |\vec{V}||\vec{W}| \cos\theta \tag{33}$$

Cross Product

$$\vec{V} \times \vec{W} := \langle V_y W_z - W_y V_z, V_x W_z + W_x V_z, V_x W_y - W_x V_y \rangle \tag{34}$$

The trick:

$$\vec{V}\times\vec{W} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ V_x & V_y & V_z \\ W_x & W_y & W_z \end{vmatrix} = V_yW_z - W_yV_z\hat{i} - V_xW_z + W_xV_z\hat{j} + V_xW_y - W_xV_y\hat{k} \tag{35}$$

- This defines a new vector $\vec{V}\times\vec{W}$ that is normal/orthogonal to the plane formed by the two vectors $\vec{V}$ and $\vec{W}$.
- Note to self: draw visualization of cross product on board and discuss the right hand rule.
- Note the commutitivity of the dot product (because it results in a scalar):

$$\vec{V}\cdot\vec{W} = \vec{W}\cdot,\vec{V} \tag{36}$$

and the anticommutivity of the cross product (because it results in a vector, which carries directionality):

$$\vec{V}\times\vec{W} = -\vec{W}\cdot\vec{V} \tag{37}$$

Activity 9. *Compute the **dot and cross** product for the following pairs of vectors:*

1. $\vec{V} = \langle\alpha,\beta,\mu\rangle$ & $\vec{W} = \langle-\alpha,-\beta,-\mu\rangle$
2. $\vec{V} = \langle\alpha,0,0\rangle$ & $\vec{W} = \langle\alpha,0,0\rangle$
3. $\vec{V} = \langle\alpha,0,0\rangle$ & $\vec{W} = \langle0,\alpha,0\rangle$

1.6.2 Basis Vectors

- Basis vectors are a incredibly important set of vectors, especially in our study of general relativity
- Basis vectors are a set $\{e_i\}$ such that
  1. Any vector $\vec{V}$ can be constructed as a linear combination of these vectors
  2. These vectors are linearly independent
     - You cannot construct one as a combination of the others

A simple example in $\mathbb{R}^2$ is

$$\vec{e}_1 = \langle 1, 0\rangle \ \& \ \vec{e}_2 = \langle 0, 1\rangle \tag{38}$$

because any vector $\vec{V} = \langle x, y\rangle$in $\mathbb{R}^2$ can be written as

$$\vec{V} = x\vec{e}_1 + y\vec{e}_2 \tag{39}$$

Note that $\mathbb{R}^2$ is the set

$$\mathbb{R}^2 = \{\langle x, y\rangle : x \in \mathbb{R}, y \in \mathbb{R}\} \tag{40}$$

1.6.3 Orthonormal Basis Vectors

- *Orthonormal* basis vectors are a special set of basis vectors $\{\vec{e}_i\}$, which have unit length $|\vec{e}| = 1$ and which are constructed to all be orthogonal to each other:

  $$\hat{e}_i \cdot \hat{e}_j = \delta_{ij} \tag{41}$$

  Notice that I've begun using $\hat{e}$ rather than $\vec{e}$ to indicate that these vectors are unit length.

- Most intuitive and important example is Cartesian basis vectors in $\mathbb{R}^3$:

  $$\begin{aligned} \hat{e}_1 &= \langle 1, 0, 0\rangle \\ \hat{e}_2 &= \langle 0, 1, 0\rangle \\ \hat{e}_3 &= \langle 0, 0, 1\rangle \end{aligned} \tag{42}$$

- Note that another hugely important orthonormal set of basis vectors is the trigonometric functions $\sin(nx)$ and $\cos(nx)$ on the interval $[-\pi, \pi]$, though there are some more details needed to show this is true.

1.6.4 Matrices

- Matrices are one of the most important mathematical objects, and will be especially important to our later discussion of modern gravity

- Definition: A matrix is a collection of numbers that is subject to standard algebraic operations:

  $$A = \begin{pmatrix} a_{11} & a_{12} & \dots a_{1n} \\ a_{21} & a_{22} & \dots a_{2n} \\ \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots a_{mn} \end{pmatrix} \tag{43}$$

A is called an $(m \times n)$ matrix: m rows and n columns.

Question 10. *What two algebraic operations do they think act on matrices?*

Answer: addition and multiplication. Any ideas for how matrices are added?

- **Addition** of matrices:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix} \tag{44}$$

- **Multiplication** of matrices:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae+bg & af+bh \\ ec+gd & fc+hd \end{pmatrix} \tag{45}$$

- Notice that matrix multiplication need not always be done by matrices of the same size:

$$\begin{pmatrix} \\ A \\ \\ \end{pmatrix}_{m_1 \times n_1} \begin{pmatrix} \\ B \\ \\ \end{pmatrix}_{m_2 \times n_2} = \begin{pmatrix} \\ C \\ \\ \end{pmatrix}_{m_1 \times m_2} \tag{46}$$

- If $n_1 = n_2$, $A \cdot B$ is an allowed operation. Notice that multiplication when $m_1 \neq m_2$ is not allowed.

Activity 10. *Multiply the following matrices:*

1.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{2\times 2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{2\times 2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{2\times 2} \tag{47}$$

2.

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}_{2\times 2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}_{2\times 2} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}_{2\times 2} \tag{48}$$

3.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{2\times 2} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}_{2\times 1} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}_{2\times 1} \tag{49}$$

The last example motivates a new representation of vectors.

- Vectors can be represented as **column matrices**.

$$\vec{V} = \langle V_x, V_y, V_z \rangle \equiv \begin{pmatrix} V_x \\ V_y \\ V_z \end{pmatrix}_{3\times 1} \tag{50}$$

In doing this, we can transform them under matrix multiplication, as in the last example.

- Under the column vector representation of matrices, we define the transpose operator as

$$\vec{V}^T := \begin{pmatrix} V_x & V_y & V_z \end{pmatrix}_{1\times 3} \tag{51}$$

which turns our column vector into a **row vector**.

- Under these representations, we have a new way of representing dot products (any thoughts for how the following matrix multiplication will be carried out?):

$$\vec{V} \cdot \vec{W} = \vec{V}^T \vec{W} = \begin{pmatrix} V_1 & V_2 & V_3 \end{pmatrix} \begin{pmatrix} W_1 \\ W_2 \\ W_3 \end{pmatrix} \tag{52}$$

1.6.5 Summary

1. Dot products:

$$\begin{gathered} (\cdot) : \vec{V}, \vec{W} \to c \quad \text{under} \\ \vec{V} \cdot \vec{W} = \sum_i V_i W_i = \begin{pmatrix} V_1 & V_2 & V_3 \end{pmatrix} \begin{pmatrix} W_1 \\ W_2 \\ W_3 \end{pmatrix} \end{gathered} \tag{53}$$

2. Cross products:

$$\begin{gathered} (\times) : \vec{V}, \vec{W} \to \vec{Z} \quad \text{under} \\ \vec{V} \times \vec{W} = \ldots \end{gathered} \tag{54}$$

Question 11. *Can we think of a way to construct an operation that takes two vectors to a* matrix *using our new column and row matrix representation of vectors.*

- *Answer*: the **outer product** $\otimes$:

$$\vec{V} \otimes \vec{W} = VW^T = \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} \begin{pmatrix} W_1 & W_2 & W_3 \end{pmatrix} \tag{55}$$

- To begin moving towards general relativity, we define the central object; the **metric tensor**:

$$g_{\mu\nu} = \hat{e}_\mu \otimes \hat{e}_\nu \tag{56}$$

  which tells us how to compute lengths and angles local to a point (which will eventually become the object that describes the curvature of spacetime)

2 Day 2: General Relativity

2.1 Lorentz Transformations in Matrix Form

Recall the Lorentz transformation

$$\begin{cases} t' &= \gamma\big(t - \frac{vx}{c^2}\big) \\ x' &= \gamma(x - vt) \\ y' &= y \\ z' &= z \end{cases} \tag{57}$$

We can make these expressions much simpler if we define $\beta := v/c$ and absorb a factor of c into t and t' (notice this also gives units of length). This results in

$$\begin{cases} ct' &= \gamma\big(ct - \beta x\big) \\ x' &= \gamma(x - \beta ct) \\ y' &= y \\ z' &= z \end{cases} \tag{58}$$

Question 12. *Knowing what we do about matrix multiplication, can we now discover a matrix that, when applied to the column vector*

$$x^\mu = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}, \tag{59}$$

results in the correct expression for the Lorentz transformation?

Answer:

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} \tag{60}$$

- We have therefore discovered the matrix form of the Lorentz transformation

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{61}$$

- To make four dimensional algebra work, either the zeroth term (ct component) must carry a negative sign or all of the spatial components must carry it. These are called metric signatures. We will adapt $(-+++)$, which is oddly used more frequently on the east coast, and so is called the east coast convention.

- Notice we have introduced the notation

$$x^\mu = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} \tag{62}$$

  Where $\mu = 0$ corresponds to the ct component and so on. This is called **Einstein notation**

- In Einstein notation, the row matrices are represented with its indices "downstairs":

$$x_\mu = \begin{pmatrix} -ct & x & y & z \end{pmatrix} \tag{63}$$

- The x_μ is called the **covariant** form and x^μ is called the **contravariant** form. The covariant form carries the negative sign on the ct term.

- In Einstein notation, sums over repeated indices is implied:

$$ds^2 = dx_\mu dx^\mu \tag{64}$$

  Question 13. *Can anyone recognize what the form of ds^2 looks like when*

$$dx^\mu = \begin{pmatrix} cdt \\ dx \\ dy \\ dz \end{pmatrix}? \tag{65}$$

  What does the multiplication of $dx_\mu dx^\mu$ look like?

  Answer:

$$ds^2 = \begin{pmatrix} -cdt & dx & dy & dz \end{pmatrix} \begin{pmatrix} cdt \\ dx \\ dy \\ dz \end{pmatrix} = -(cdt)^2 + (dx)^2 + (dy)^2 + (dz)^2 \tag{66}$$

- Notice that we can use the metric to transform between covariant and contravariant forms.

  Activity 11. *If*

$$dx^\mu = \begin{pmatrix} cdt \\ dx \\ dy \\ dz \end{pmatrix} \quad \& \quad dx_\mu = \begin{pmatrix} -cdt & dx & dy & dz \end{pmatrix}, \tag{67}$$

  What 4x4 matrix, when applied to dx^μ, would produce to dx_μ?

  Answer:

$$\Lambda = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{68}$$

  This result is called the **Minkowski metric**, which is denoted as $g_{\mu\nu} = \eta_{\mu\nu}$ since it is the metric used in special relativity.

  Question 14. *$\eta^{\mu\nu}$ is the inverse of $\eta_{\mu\nu}$.*

  Hint: *To take an inverse of a matrix, you replace its rows and columns so that*

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \tag{69}$$

  becomes

$$A^{-1} = \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix} \tag{70}$$

  Answer: $\eta_{\mu\nu} = \eta^{\mu\nu}$.

- The object ds^2 is called the **spacetime interval**, which tells you how distances are measured in curved spacetimes since it takes the geometry of your space into account with the $g_{\mu\nu}$ term.
  - Remember that $g_{\mu\nu}$ takes the outer product of basis vectors; $g_{\mu\nu} = \hat{e}_\mu \otimes \hat{e}_\nu$.

Question 15. *If $\eta_{\mu\nu}$ represents spacetime in special relativity, what sort of spacetime is being described in special relativity?*

Hint: *Think about the purely spatial terms $dx^2 + dy^2 + dz^2$... this is clearly space that we are used to dealing with*

Answer: *The Minkowski metric describes **flat spacetime!** Special relativity deals with environments with no curvature in spacetime.*

- Important to emphasize: A Lorentz transformation is *not* a spacetime metric. A Lorentz transformation transfers you between reference frames, while metrics tell you what spacetime looks like at a point.
  - Lorentz transformations can therefore be applied to metrics to tell you how spacetime looks in different reference frames:

$$g_{\alpha'\beta'} = \Lambda^{\mu}_{\alpha'}\Lambda^{\nu}_{\beta'}g_{\mu\nu} \tag{71}$$

- We are starting to get to the point where we want to describe the curvature of spacetime. As we will learn, Einstein discovered that the curvature of this object called spacetime is responsible for gravity.
- To begin dealing with gravity, we should first review the most basic theory of gravity

2.2 Newtonian Gravity

- Second Law

$$F = ma \tag{72}$$

- Newton realizes that

$$F_g = \frac{GMm}{r^2} \tag{73}$$

derives Kepler's laws.

Question 16. *What form do you get for a when just the gravitational force is considered?*

Answer:

$$ma = \frac{GMm}{r^2} \implies a = \frac{GM}{r^2} \tag{74}$$

- Kepler's laws had been formulated 68 years prior to Newton's realization, but had only been discovered *empirically*; only discovered by analyzing observational planetary data from Tycho Brahe (this was done by trail-and-error fitting)

- Newton was the first to *derive* Kepler's laws from theoretical principles by using Eq. ref (73)

  Activity 12. *If time and interest allows, here is how one of Kepler's laws is derived from Newtonian gravity:*

  Using

  $$a = \frac{GM}{r^2} \tag{75}$$

  and the known centripetal acceleration

  $$a = \frac{v^2}{r}, \tag{76}$$

  we get

  $$\frac{v^2}{r} = \frac{GM}{r^2} \implies v^2 = \frac{GM}{r}. \tag{77}$$

  Assuming a circular orbit, we know a planet orbits 2π radians over its total period T, which means our usual expression for velocity, $\Delta x/\Delta t$, becomes

  $$v = \frac{2\pi r}{T} \implies v^2 = \frac{4\pi^2 r^2}{T^2}. \tag{78}$$

  Setting these expressions equal to each other:

  $$\frac{GM}{r} = \frac{4\pi^2 r^2}{T^2} \implies T^2 = \frac{4\pi^2}{GM} r^3 \tag{79}$$

  This is Kepler's third law, which tells us that the cube of a planet's radius will always be related to the square of its orbital period by a constant $4\pi^2/GM$

- Newtonian gravity therefore became the reigning gravitational paradigm for centuries

- Problems with Newtonian gravity:

  1. Force is instantaneous

2. Unable to explain some observational phenomena: perihelion of Mercury
3. No *why* gravity

- Aside:
  - $1/r^2$ behavior in Newtonian gravity was not original at the time of Newton
  - Inverse square law behavior is ubiquitous in physics due to the surface area of a sphere $SA = 4\pi r^2$

2.3 Transition to General Relativity

- The **equivalence principle**:
  - Einstein realized that freely falling was equivalent to being subject to a gravitational field. An unaware observer would not be able to tell the difference
  - Gravity is therefore equivalent to acceleration
- It is therefore implied that gravity is *not* a strict force as Newton saw it, but a result of reference frames, just as

2.4 General Relativity

Question 17. *If gravity is the curvature of spacetime under general relativity, how can we describe the curvature of spacetime?*

Hint: *how is curvature typically defined for regular functions? (derivatives) What is the object that defines spacetime? ($g_{\mu\nu}$)*

Answer: *We need a way of defining derivatives of the metric.*

- Taking derivatives of the metric is much more complicated than doing this for simple functions, however, since the metric is a 4x4 matrix.
  - This is much like how derivatives get more complicated in 3D because you now have more degrees of freedom for your function to change in

- There are entire classes dedicated to defining derivatives in curved spacetimes, and this is indeed what the first ∼half of a course on general relativity is dedicated to.

- It is therefore in the interest of this directed study to flash the field equations and explain what the terms represent, rather than building them from the ground up, as Einstein did (it took him ten years after all)

- The Einstein field equations, which relate the curvature of spacetime to the distribution of matter in it (which describes gravity), are

$$\boxed{R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}} \tag{80}$$

  The left hand side is typically shortened to one object, called the Einstein tensor:

$$G_{\mu\nu} := R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} \tag{81}$$

- The exact form of $R_{\mu\nu}$, the Ricci tensor, is

$$R_{\mu\nu} = \partial_\alpha \Gamma^\alpha_{\mu\nu} - \partial_\nu \Gamma^\alpha_{\mu\alpha} + \Gamma^\alpha_{\mu\nu}\Gamma^\beta_{\alpha\beta} - \Gamma^\alpha_{\mu\beta}\Gamma^\beta_{\nu\alpha} \tag{82}$$

  and

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2}g^{\beta\alpha}\big(\partial_\nu g_{\beta\mu} + \partial_\mu g_{\beta\nu} - \partial_\beta g_{\mu\nu}\big) \tag{83}$$

  ... this is all to say that the description of spacetime curvature contains *many* partial derivatives of the metric and the field equations are *very* difficult to solve analytically.

- What *is* important to note about the field equations is that, *typically*, the **free variable is your stress-energy tensor, and your output is the metric**, which describes the curvature of spacetime

  - You typically define the stress-energy tensor, and the field equations tell you how the resulting spacetime curvature looks
  - Though there are times you know the metric tensor for an object, and can produce the stress-energy tensor of that object as a result, such as in the case of simple black holes (typically just for pedagogical examples, however)

– *"Matter tells spacetime how to curve, and curved spacetime tells matter how to move" (John Wheeler)*

Question 18. *What are the units of the field equations and what is the value of the constant κ outside of the stress-energy tensor?*

Answer:

$$\kappa = \frac{8\pi G}{c^4} = \frac{8\pi(6.67\times 10^{-11})}{(3\times 10^8)^4}\left[\frac{m^3}{kg\ s^2}\left(\frac{s}{m}\right)^4\right] \approx 2\times 10^{-43}\left[\frac{s^2}{kg\ m}\right] \quad (84)$$

- $T_{\mu\nu}$ is the component of the field equations that describes the distribution of matter, which is called the stress-energy tensor. As the name implies, it carries units of stress, which is $\left[\frac{kg}{m\ s^2}\right]$.
  - This means that the units on the Einstein field equations are

$$\kappa T_{\mu\nu} = \left[\frac{s^2}{kg\ m}\right]\left[\frac{kg}{m\ s^2}\right] = \left[\frac{1}{m^2}\right] \quad (85)$$

- This makes sense when you consider the curvature of spacetime is the quantity being described
- Definition of the stress-energy tensor:

  $T_{\mu\nu}$ **is the** $\mu-$**momentum across a surface of constant** dx_ν**.**

  By $\mu-$momentum, we refer to four-momentum, which is defined here:

$$p_\mu = \langle \frac{E}{c}, p_i \rangle \quad (86)$$

  This results in

$$\begin{aligned} T_{00}&: \text{ Energy density} \\ T_{0i}&: \text{ Energy flux across } dx_i \text{ surface} \\ T_{i0}&: \ i\text{-Momentum density} \\ T_{ij}&: \ i\text{-Momentum flux across } dx_i \text{ surface} \end{aligned} \quad (87)$$

  Where momentum density refers to the transfer of momentum across a unit area
  - This is why $T_{\mu\nu}$ represents physical objects in a spacetime.

2.5 Schwarzschild Metric

- The existence of black holes comes directly from the Einstein field equations when you input a vacuum stress-energy tensor: $T_{\mu\nu} = 0$
- Doing so results in the metric:

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2GM}{c^2 r}\right) & 0 & 0 & 0 \\ 0 & \frac{1}{1-\frac{2GM}{c^2 r}} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2\theta \end{pmatrix} \tag{88}$$

- This metric was first discovered by Karl Schwarzschild in 1915, so it is called the Schwarzschild metric
  - Notice that this is in spherical coordinates: (t, r, θ, ϕ) (draw this out on the board... θ is radial and ϕ is azimuthal ($\theta \in [0, 2\pi]$))
  - Also notice that this assumes the black hole is
- To see why this is a black hole, consider the spacetime interval

$$ds^2 = -\left(1 - \frac{2GM}{c^2 r}\right)c^2 dt^2 + \frac{dr^2}{1 - \frac{2GM}{c^2 r}} + r^2\left(d\theta^2 + \sin^2\theta d\phi^2\right) \tag{89}$$

Question 19. *What happens to $g_{\mu\nu}$ when:*

1. $r \to \infty$
2. $M = 0$
3. $r = \frac{2GM}{c^2}$

- The last part of this question tells us that any object whose mass is compressed to a radius smaller than $r = \frac{2GM}{c^2}$ will form a black hole

3 Day 3: Applications of and Problems with General Relativity

Topics to be covered in slide presentations and their outlines:

1. **Cosmology**:
    - Discuss the FRLW metric and talk about the resulting three possible geometries of the universe. Also discuss the fact that a flat universe ($k = 0$) fits observational data
    - Inflationary cosmology:
        - Talk about how the standard big bang picture is not the end of the story
        - Talk about the varying kinds of inflationary cosmology and their philosophical implications
    - The CMB:
        - What it is, how it formed, and why it's such an important source of observational data
        - Talk about the gaussian nature of the fluctuations and how non-gaussianities could probe the early universe
2. **Gravitational waves**:
    - Discuss the history of gravitational wave physics and the current state of LIGO + what LIGO has helped us discover about the universe
    - Talk about the relevant funding cuts to the NSF which will have an outsized impact on LIGO
3. **Problems** with general relativity and some of their potential solutions:
    - Dark matter:
        - Get into the discrepancies between predictions generated by general relativity and observational data that lead to dark matter
        - Discuss the landscape of dark matter theories (emphasize that there is not *one* theory of dark matter)

  - Talk about the work in modified gravity as well
- Dark energy:
  - The mysterious Λ factor and Einstein's famous "blunder" in adding it to the field equations
  - Observational evidence for its existence and consequences of this
- Singularities:
  - Discuss how these come up at the center of black holes and at the origin of the universe
  - Talk about why quantum gravity becomes relevant at these singularities and how theories of quantum gravity are being developed to attempt solving these problems

4. **Research in relativity**:

- Talk about what sort of work you have done in the field
- Talk about problems that people have worked on recently in the field and those that are still open